北京郵電大學

Queen Mary University of London

For examiners' use only

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EBU6501 A

Joint Programme Examinations 2023/24

EBU6501 Middleware

Paper A

Time allowed 2 hours

Answer ALL Questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Gokop Goteng, Dr Ethan Lau

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) State the type of Publish-Subscribe messaging-based Message-oriented Middleware (MOM) model in Figure 1 below and explain the usage of the model.

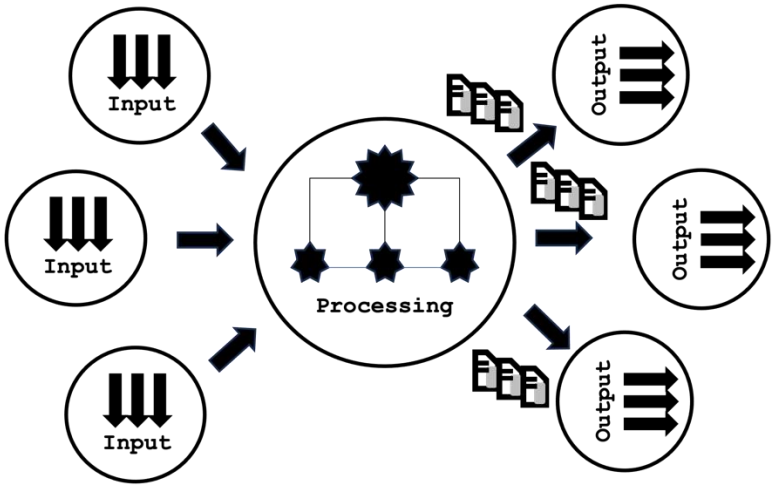


Figure 1.

(5 marks)

	Do not write in this column
	5 marks

b) Explain the role of the object **`MessageBroker`** in the context of MOM architecture. What is the main object component inside the **`MessageBroker`**?

(4 marks)

	Do not write in this column

		4 marks

c) The Figure 2 below demonstrates an example of the Java Message Service (JMS) API Programming model.

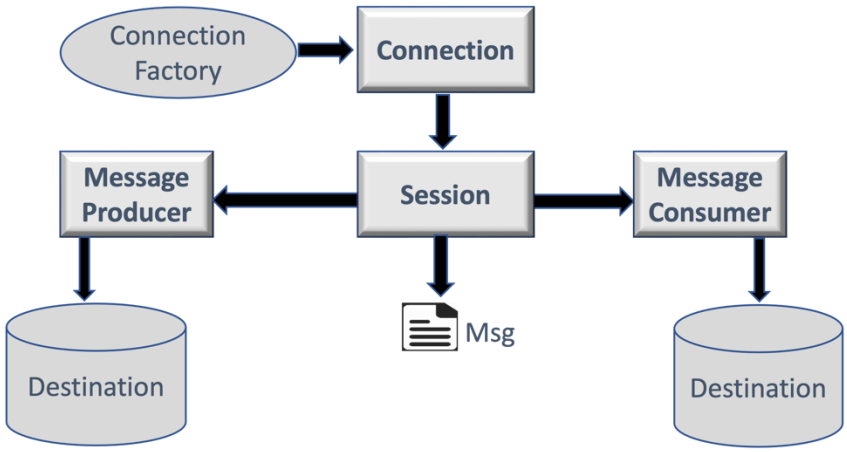


Figure 2.

(8 marks)

i) Describe the TWO types of the JMS Message consumption from the **`Destination`** object.

[6 marks]

ii) Given that the connection is from Figure 2 is already established through the object **`TopicConnection`**, create a **`session`** object through **`TopicSession.`** The transaction is set to **`false`** and the acknowledgement mode is set to **`AUTO_ACKNOWLEDGE`**.

[2 marks]

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		8 marks

d) Figure 3 below shows the directory structure of the Tomcat container.

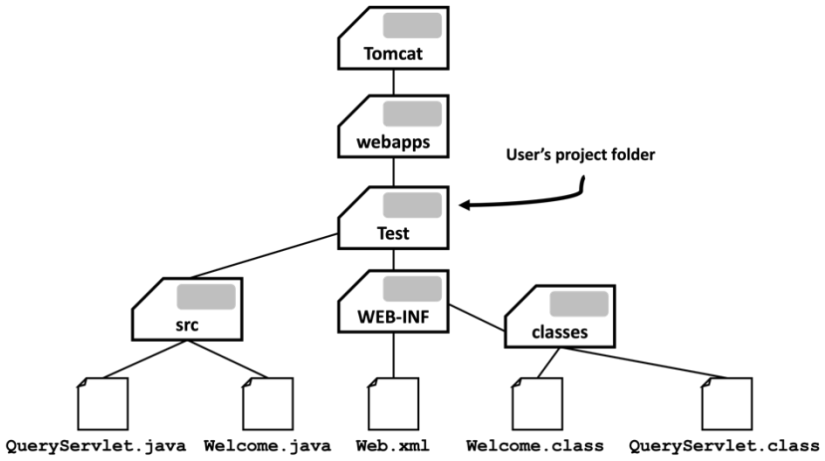


Figure 3.

(8 marks)

i) Carefully denote the pages that are belong to the individual design pattern of the ***Model***, ***View*** and ***Controller*** model (write down "***n/a"*** or "***none***" if there is/are none).

[3 marks]

ii) Name the directory file that corresponds to the Deployment Descriptor in Figure 3. Explain why it is located under the folder directory**.**

[3 marks]

iii) Write the required **`service()`** method in Java language that called by the server to allow a servlet to handle a POST request. Includes the necessary exception handlers.

[2 marks]

	Do not write in this column
	8 marks

Question marking $= \frac{}{5} + \frac{}{4} + \frac{}{8} + \frac{}{8} = \frac{}{25}$

Question 2

a) Spring Framework is an application framework in software engineering, and it is most often used in the context of object-oriented programming such as Java.

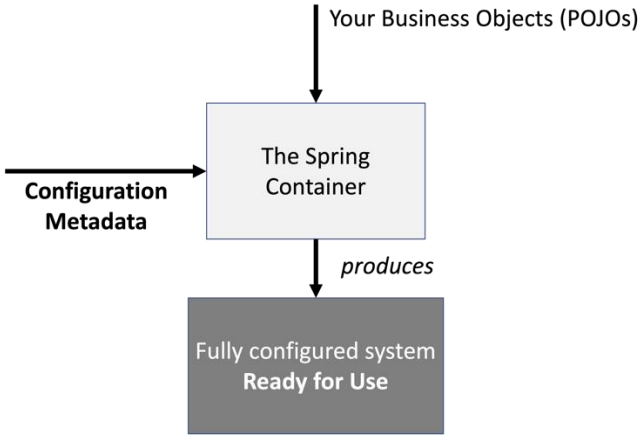


Figure 4.

[9 marks]

i) Based on the information in Figure 4, what is the core principle of technology used in the Spring Container? Explain its role in the Spring Framework.

(3 marks)

ii) List out the TWO examples of the Spring Container.

(2 marks)

iii) The Application Context is Spring's advanced container. Implement the Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "**login.xml**". Finally, obtain the bean with the bean ID = **`Welcome`** from the **`Welcome`** class.

(4 marks)

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	9 marks

b) Spring Aspect-oriented programming (AOP) is a technique for building common but reusable routines that can be applied in applications widely.

Figures 5a – c show Java codes with modularisation of concerns through **`TrackOperation.java`** at the source codes **`MainApp.java`** and **`Operation.java`**. Predict the output of the program while maintaining the order of execution.

```
public class MainApp {

    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");  Operation op =
        (Operation) context.getBean("opBean");
        System.out.println("Calling Operation A...");
        op.a();
        System.out.println("Calling Operation B...");
        op.b();
```

Figure 5a.

```
public class TrackOperation { // Aspect class
   @Pointcut("execution(* Operation.*(..))")
   public void pc(){}//pointcut name

   @After("pc()")//applying pointcut on after advice
   public void myadvice(JoinPoint jp)
   {
      System.out.println("Operation terminated successfully.");
```

Figure 5b.

```
public class Operation {
   public void a(){System.out.println("Operation A completed.");}
   public int b(){System.out.println("Operation B completed.");}}
```

Figure 5c.

(4 marks)

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	4 marks

c) This question is about Java threads.

(12 marks)

i) Suggest the corresponding methods to avoid the ***Visibility*** and ***Race Conditions*** issues.

[2 marks]

ii) The code in Figure 6 below shows Java class **`MyAddons`** where the method **`getNextValue()`** is not thread-safe as it is accessible by multiple threads. Improve the code below ensuring the thread-safe and atomic operation.

```
public class MyAddons {
  private int value = 100;
  public int getNextValue(){
            return value++;
}
```

Figure 6.

[3 marks]

iii) Explain the usage of ***Mutual Exclusion*** and ***Cooperation*** in Java's monitor.

[3 marks]

iv) In your own words, explain the differences of **`notify()`** and **`notifyAll()`** in Java's monitor.

[4 marks]

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		12 marks

Question marking $= \frac{}{9} + \frac{}{4} + \frac{}{12} = \frac{}{25}$

Question 3

a) This question is on Apache Kafka.

[18 marks]

i) Describe how Middleware Developers use the Apache Kafka's ***Events***, ***Topics*** and ***Partitions*** to develop Message-Oriented Middleware (MOM) applications that can ingest (accept as inputs) and organise different types of data from multiple sources at large scale.

(8 marks)

ii) Describe Apache Kafka ***Brokers*** and how you can mitigate against losing data in case of any failure in the cluster.

(2 marks)

iii) Figure 7 below is a Kafka Consumer Library written in Java programming language. Write the corresponding Producer Library using the Java class KafkaProducer and Java collection ProducerRecord in Apache Kafka.

(8 marks)

```
 try (final KafkaConsumer<String, Payment> consumer = new
KafkaConsumer<>(props))
     {consumer.subscribe(Collections.singletonList(TOPIC));

     while (true) {
        ConsumerRecords<String, Payment> records = consumer.poll(100);
        for (ConsumerRecord<String, Payment> record : records) {
            String key = record.key();
            Payment value = record.value();
            System.out.printf("key = %s, value = %s%n", key, value);
        }
      }
    }
```

Figure 7: Kafka Consumer Library

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	18 marks

b) Describe ***Symmetric*** and ***Asymmetric*** Encryption methods.

[5 marks]

	Do not write in this column
	5 marks

c) Describe TWO IoT protocols that are commonly used in buildings and home automation applications.

[2 marks]

	Do not write in this column
	2 marks

Question marking: $\frac{}{18} + \frac{}{5} + \frac{}{2} = \frac{}{25}$

Question 4

a) Using your knowledge of the Open Services Gateway Initiatives (OSGi) architecture, describe how ***Modules***, ***Services Registry*** and ***Execution Environment*** work in OSGi.

[6 marks]

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	6 marks

b) A company wants to deploy message-oriented middleware (MOM) in its cloud platform that will ensure that its architecture is loosely coupled and can satisfy both push model (passive model) and pull model (poll or active model) applications. Describe TWO cloud services that can help the company to implement loosely coupled architecture with push and pull models.

[8 marks]

	Do not write in this column

	8 marks

c) Write a public client Java class program named NetworkClient that makes request to a server with IP address 100.0.1.0 at port 8080. Use the Java Socket class to make the connection and to read data using BufferedReader and InputStreamReader Java classes.

[11 marks]

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		11 marks

Question marking $\frac{}{6} + \frac{}{8} + \frac{}{11} = \frac{}{25}$

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2023-2024

Rough Working

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